

3(a)(i)	angle (subtended) where arc (length) is equal to radius	M1
	(angle subtended) at the centre of a circle	A1
3(a)(ii)	angular frequency = $2\pi \times \text{frequency}$ or $2\pi / \text{period}$	B1
3(b)(i)	c/ML^3 is a constant so acceleration is proportional to displacement	B1
	minus sign shows that acceleration and displacement are in opposite <u>directions</u>	B1
3(b)(ii)	$c/ML^3 = (2\pi f)^2$	C1
	$c = 4\pi^2 \times 3.2^2 \times 0.24 \times 0.65^3$	C1
	$= 27 \text{ kg m}^3 \text{ s}^{-2}$	A1

Question	Answer	Marks
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